Analysis of Variance (ANOVA)

Dr Phikolomzi Matikinca

Department of Environmental and Geographical Science (EGS)
University of Cape Town (UCT)
Office: 4.08
Email: phikolomzi.matikinca@uct.ac.za

Analysis of Variance (ANOVA)

- If we obtain measurements from 2 independent samples = **independent samples t-test**

- However, we often obtain data for 3 or more samples - requires a multi-sample analysis such as ANOVA

- **ANOVA tests for differences between ≥ 3 groups**

What does ANOVA do?

- Tests the effects of **one or more discrete independent variable/s** on a single dependent variable

- Independent variables ~ **factors**
  - Each factor has 3 or more groups (treatments or **levels** of the factor)
  - Usually nominal or ordinal variables (with categorical groups)

- Dependent variables ~ **response** variables
  - What is measured
  - Can only be ratio or interval data

Example

- Is **tadpole survival** affected by pesticide concentration? Set up an experiment with three pesticide treatments classified as **low** (0 mg/L), **medium** (1 mg/L) and **high** (2 mg/L).

- The factor is **pesticide concentration** with 3 treatments or groups.
- The dependent variable is **tadpole survival**

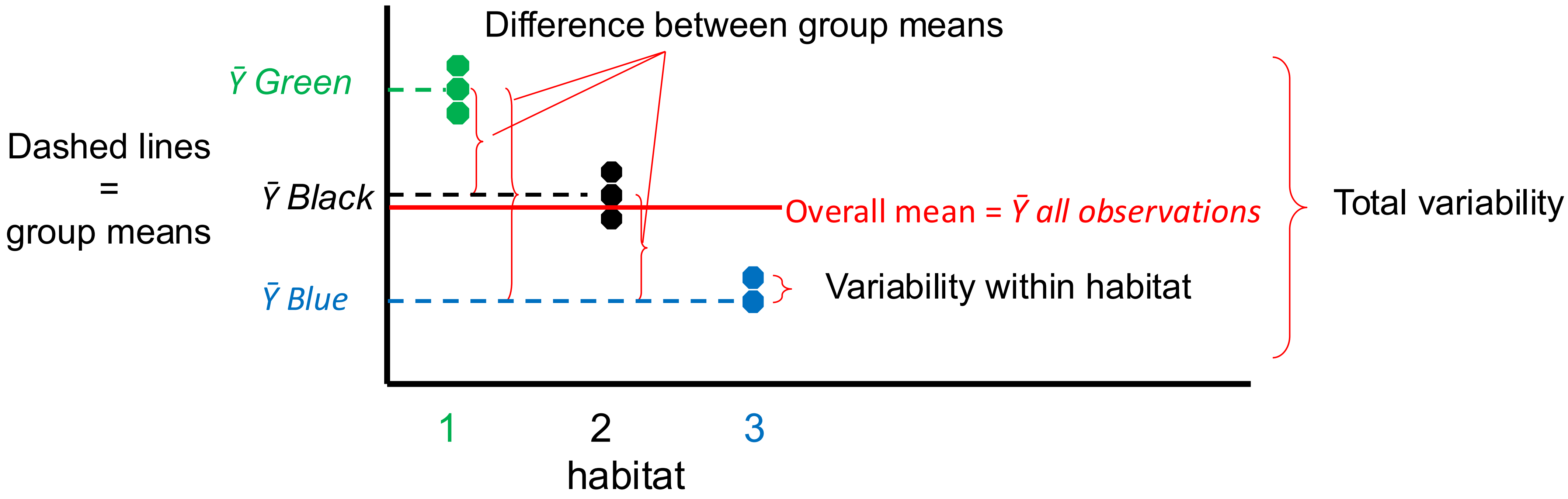
- In this case, pesticide concentration is an **ordinal** variable with 3 discrete, categorical groups (low, med, high).

What does ANOVA do?

Remember…when considering variability in the data

- there is variation within a dataset (think of the frequency distribution of data)
- and there is variation between datasets (the reason you test for differences between samples)
- **ANOVA** attempts to partition the total variability (SST) of the dependant variable into:
  - variability related to the factors investigated in the study (**factors / predictors**)
  - variability due to something else (experimental error or unexplained variation, random variation, residual variation (SSE))
  - Calculates the ratio of these two sources of variability:
    - if the variability related to the factor is higher than unexplained variability, then variability in the dataset is due to the factors or treatment (group) levels
    - Therefore, null hypothesis of equal means is rejected

Variability in data



If variability due to the factor is **high** relative to variability within groups, then variation in the data is likely due to the effect of the factor being tested

If variability due to the factor is **low** relative to variability within groups, then variation in the data is likely due to something other than the factor (unexplained)

One-way ANOVA

- Single factor ANOVA
- Tests the effects of **ONE** factor with **≥ 3** groups on the dependent variable
- Tests for differences between group means
- [Direct extension of the independent-samples t-test, which examines the effect of one independent variable with only 2 groups]
- Used in Completely Randomized Design (CRD)

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

One-way ANOVA

- Remember that ANOVA calculates a ratio of the two sources of variability
- Specifically, it calculates a ratio of variance
- Sample variance:

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

- To test H_0 of no difference between group means, the ANOVA or F statistic is calculated:

$$F = \frac{\text{MS Groups}}{\text{MS Error}}$$

Variability due to factor

Unexplained variability

Sources of variation

	Source of Variation	Sum of Squares (*SS*)	Degrees of Freedom (***df***)	Mean square (*MS*), *or* **Variance** (*s*)	***F***
	Total (overall variance)	$\sum_{i=1}^{k}\sum_{j=1}^{n_i}(X_{ij}-\bar{X}_i)^2$	*n - 1*	$\frac{SS_{Total}}{df_{Total}}$	
Variability due to treatment	Groups (between group variance)	$\sum_{i=1}^{k} n_i\left(\bar{X}_i-\bar{X}\right)^2$	*k -* 1	$\frac{SS_{Groups}}{df_{Groups}}$	$\frac{MS_{Groups}}{MS_{Error}}$
Unexplained variability	Error (within group variance)	$\sum_{i=1}^{k}\sum_{j=1}^{n_i}(X_{ij}-\bar{X})^2$	*n - k*	$\frac{SS_{Error}}{df_{Error}}$	

$\left(\bar{X}_i-\bar{X}\right)$ deviation between the mean of the i^{th} treatment and the overall mean

$\left(X_{ij}-\bar{X}_i\right)$ deviation between an observation (j^{th}) of the treatment (i^{th}) and the mean of observations in the same treatment (i^{th})

$(X_{ij}-\bar{X})$ deviation between an observation (j^{th}) of the treatment (i^{th}) and the overall mean

n = total number of all replicates
k = number of groups (levels, treatments)
n_i = number of replicates in each group

Calculating One-way ANOVA

Example. In a vegetation management study of a landscape, plots were subjected to one of three experimental treatments (groups): burned, tilled soil, or no treatment (control).

Twenty-one plots were randomly assigned to the treatments, with seven plots per treatment. After two years, the species richness of the vegetation was determined for each plot.

Is the species richness of vegetation different among the three treatments?

Calculating One-way ANOVA

Aim: To determine the effect of different management strategies on grassland vegetation species richness.

Objective: To apply three different experimental treatments (burning, tilling and no treatment) to randomly assigned grassland plots.

Hypotheses:

H_0: the type of management strategy used will have **NO** effect on vegetation species richness.

H_A: the type of management strategy used will have **different effects** on vegetation species richness.

Calculating One-way ANOVA

Step 1. Determine the variability due to the **factor** (treatment levels of the factor):

- Calculate SS_{groups} i.e. the SS (sum of the squared deviations) of each group mean from the overall mean

Groups (between group variance	$\sum_{i=1}^{k} n_i \left(\overline{X}_i - \overline{X}\right)^2$

	Species Richness			Deviation from Treatment Mean $(X_{ij} - \bar{X}_i)$			
	Control	Burned	Tilled	Control	Burned	Tilled		
	3	6	6	-1.7	-1.7	0.4	
	6	7	5	1.3	-0.7	-0.6	
	2	10	6	-2.7	2.3	0.4	
	5	7	8	0.3	-0.7	2.4	
	7	9	5	2.3	1.3	-0.6	
	6	8	5	1.3	0.3	-0.6	
	4	7	4	-0.7	-0.7	-1.6	
Sample size	7	7	7				
Treatment mean: $\bar{X}_i =$	4.7	7.7	5.6				**Overall mean:** $\bar{X} =$ 6
$n_i =$	7	7	7				$n =$ 21
							$k =$ 3

	$(\bar{X}_i - \bar{X})$	$(\bar{X}_i - \bar{X})^2$	$n_i(\bar{X}_i - \bar{X})^2$
Control:	4.7 -6 = -1.3	$(-1.3)^2$ = 1.69	(7*1.69) = 11.83
Burned:	7.7 – 6 = 1.7	$(1.7)^2$ = 2.89	(7*2.89) = 20.23
Tilled:	5.6 – 6 = -0.4	$(-0.4)^2$ = 0.16	(7*0.16) = 1.12

$$\sum_{i=1}^{k} n_i(\bar{X}_i - \bar{X})^2$$

TOTAL = 33.18

Calculating One-way ANOVA

- $\therefore SS_{groups} = 33.18$
- Next, calculate MS_{groups} i.e. the mean square or variance between groups

$$\frac{SS_{Groups}}{df_{Groups}}$$

  - $Df_{groups} = k - 1 = 2$
  - $SS_{groups} = 33.18$

- **$\therefore$ MS_{Groups} = 16.59**

Calculating One-way ANOVA

Step 2. To determine the unexplained variability ("error")

- Calculate SS_{error} i.e. the sum of the SS of the observations in each group from the group mean

$$\sum_{i=1}^{k}\sum_{j=1}^{n_i}(X_{ij}-\overline{X}_i)^2$$

Species Richness			Deviation from Treatment Mean $\left(X_{ij} - \overline{X}_i\right)$		
Control	Burned	Tilled	Control	Burned	Tilled
3	6	6	-1.7	-1.7	0.4
6	7	5	1.3	-0.7	-0.6
2	10	6	-2.7	2.3	0.4
5	7	8	0.3	-0.7	2.4
7	9	5	2.3	1.3	-0.6
6	8	5	1.3	0.3	-0.6
4	7	4	-0.7	-0.7	-1.6

$$\sum_{j=1}^{7}\left(x_{ij} - \overline{x}_i\right)^2 =$$

19.43	11.43	9.72

$$\sum_{i=1}^{k}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_i)^2$$

= 19.43 + 11.43 + 9.72
= 40.58

Calculating One-way ANOVA

- $\therefore SS_{error} = 40.58$

- Next, calculate MS_{error} i.e. the mean square or variance within groups

$$\frac{SS_{Error}}{df_{Error}}$$

  - $Df_{error} = n - k = 21\text{-}3 = 18$
  - $SS_{error} = 40.58$

- **$\therefore$ MS_{Error} = 2.254**

Calculating One-way ANOVA

Step 3. Calculate F (the ratio of these two variances)

Variance due to factor: $\mathbf{MS_{Groups} = 16.59}$

Variance that is unexplained: $\mathbf{MS_{Error} = 2.254}$

$$F = \frac{MS_{Groups}}{MS_{Error}} = \frac{16.59}{2.25} = 7.37$$

Large when the **means are different** (high variability due to the factor relative to the unexplained variability)

Small when the **means are quite similar** (small variability due to the factor relative to the unexplained variability)

Calculating One-way ANOVA

- Look up the critical F value in an F table
- Compare calculated F with critical $\mathbf{F_{0.05(2),2,18}}$

$$\mathbf{F_{0.05(2),2,18}}$$

Numerator df

Denominator df

Calculating One-way ANOVA

$$F = \frac{MS_{Groups}}{MS_{Error}}$$

numerator = df_{groups}

denominator = df_{error}

- In the critical table of F values, there are two tables for different numerator **dfs** labelled at the top of the table
- The denominator **dfs** are given down the left side of table

TABLE B.4 (cont.) CRITICAL VALUES OF THE *F* DISTRIBUTION

Numerator DF = 2

Denom. DF	α(2): 0.50 α(1): 0.25	0.20 0.10	0.10 0.05	0.05 0.025	0.02 0.01	0.01 0.005	0.005 0.0025	0.002 0.001	0.001 0.0005
1	7.50	49.5	200.	800.	5000.	20000.	80000.	500000.	2000000.
2	3.00	9.00	19.0	39.0	99.0	199.	399.	999.	2000.
3	2.28	5.46	9.55	16.0	30.8	49.8	79.9	149.	237.
4	2.00	4.32	6.94	10.6	18.0	26.3	38.0	61.2	87.4
5	1.85	3.78	5.79	8.43	13.3	18.3	25.0	37.1	49.8
6	1.76	3.46	5.14	7.26	10.9	14.5	19.1	27.0	34.8
7	1.70	3.26	4.74	6.54	9.55	12.4	15.9	21.7	27.2
8	1.66	3.11	4.46	6.06	8.65	11.0	13.9	18.5	22.7
9	1.62	3.01	4.26	5.71	8.02	10.1	12.5	16.4	19.9
10	1.60	2.92	4.10	5.46	7.56	9.43	11.6	14.9	17.9
11	1.58	2.86	3.98	5.26	7.21	8.91	10.8	13.8	16.4
12	1.56	2.81	3.89	5.10	6.93	8.51	10.3	13.0	15.3
13	1.55	2.76	3.81	4.97	6.70	8.19	9.84	12.3	14.4
14	1.53	2.73	3.74	4.86	6.51	7.92	9.47	11.8	13.7
15	1.52	2.70	3.68	4.77	6.36	7.70	9.17	11.3	13.2
16	1.51	2.67	3.63	4.69	6.23	7.51	8.92	11.0	12.7
17	1.51	2.64	3.59	4.62	6.11	7.35	8.70	10.7	12.3
18	1.50	2.62	3.55	4.56	6.01	7.21	8.51	10.4	11.9
19	1.49	2.61	3.52	4.51	5.93	7.09	8.35	10.2	11.6
20	1.49	2.59	3.49	4.46	5.85	6.99	8.21	9.95	11.4
21	1.48	2.57	3.47	4.42	5.78	6.89	8.08	9.77	11.2
22	1.48	2.56	3.44	4.38	5.72	6.81	7.96	9.61	11.0
23	1.47	2.55	3.42	4.35	5.66	6.73	7.86	9.47	10.8
24	1.47	2.54	3.40	4.32	5.61	6.66	7.77	9.34	10.6
25	1.47	2.53	3.39	4.29	5.57	6.60	7.69	9.22	10.5
26	1.46	2.52	3.37	4.27	5.53	6.54	7.61	9.12	10.3
27	1.46	2.51	3.35	4.24	5.49	6.49	7.54	9.02	10.2
28	1.46	2.50	3.34	4.22	5.45	6.44	7.48	8.93	10.1
29	1.45	2.50	3.33	4.20	5.42	6.40	7.42	8.85	9.99
30	1.45	2.49	3.32	4.18	5.39	6.35	7.36	8.77	9.90

Calculating One-way ANOVA

- Critical $\mathbf{F}_{\mathbf{0.05(2),2,18}} = 4.56$
- Calculated $\mathbf{F}_{\mathbf{0.05,2,18}} = 7.37$

- Calculated F > critical F $\therefore$ $p < 0.05$
- $\therefore$ there is a significant difference in vegetation species richness between treatments.

- Biological: different management strategies affect vegetation species richness

SPSS output for One-way ANOVA

Tests of Between-Subjects Effects

Dependent Variable: richness

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	33.429[a]	2	16.714	7.415	.004
Intercept	756.000	1	756.000	335.408	.000
treatment	33.429	2	16.714	7.415	.004
Error	40.571	18	2.254		
Total	830.000	21			
Corrected Total	74.000	20			

a. R Squared = .452 (Adjusted R Squared = .391)

Dividing **Mean Squares** of treatment (Groups) and Error gives ***F*** **= 16.714/2.254 = 7.415**

F **= 7.415** (rounding errors result in slight difference from $F = 7.37$ calculated manually, above).

p **= 0.004**

Statistical conclusion: species richness is not the same in all three treatments

Biological conclusion: burning and/or tilling have a significant effect on vegetation species richness

This is how you should describe your statistical results:

- There was a significant difference in vegetation species richness between treatments ($F_{2, 18} = 7.42$, $p = 0.004$, Table X).

A significant F value for a factor means that:

- There is a significant difference between groups means.
- <u>But</u> there are 3 groups.

- How do you determine which groups are different from each other?

We use a *Post hoc* test also called "multiple comparison tests"

- *Post hoc* means "**after the fact**" because it is only done after an ANOVA

- A *post hoc test* is only done if:
  - the effect of a factor in the ANOVA is **significant**
  - If the factor contains more than 2 levels

There are many types of *post hoc* test:

e.g.

- Least Significant Difference (LSD)
- Dunnett's
- Scheffe's
- Tukey's Honestly Significant Difference (HSD)
- Newman-Keuls
- Duncan's
- Bonferroni

There are many types of *post hoc* test:

- Pairwise comparisons between treatment levels:

	Burn	**Tilled**	**Control**
Burn		X	X
Tilled	X		X
Control	X	X	

Post hoc Tukey's test: Multiple comparison tables

Multiple Comparisons

Dependent Variable: richness
Tukey HSD

(I) treatment	(J) treatment	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
burn	control	3.00*	.802	.004	.95	5.05
	tilled	2.14*	.802	.039	.09	4.19
control	burn	-3.00*	.802	.004	-5.05	-.95
	tilled	-.86	.802	.545	-2.91	1.19
tilled	burn	-2.14*	.802	.039	-4.19	-.09
	control	.86	.802	.545	-1.19	2.91

Based on observed means.

*. The mean difference is significant at the .05 level.

The results table is set up to illustrate the results for when (I) treatment is compared with (J) treatment. In this case, "burn" is compared with "control". While it looks like there are six comparisons being made, there are only three – burn vs. control, burn vs. tilled, and control vs. tilled – but each is repeated in the table.

Mean difference = difference between the means in the two groups being compared, In this case, burn – control = 3.00. (And control – burn = -3.00).
The * next to the value indicates the difference is significant based on critical $p = 0.05$ – can be confirmed with p values that are provided in "Sig." column.

p = 0.004

Statistical conclusion: species richness in the burn treatment is different from species richness in the control

Biological conclusion: burning has an effect on vegetation species richness. In this case, burning increases species richness.

Post hoc Tukey's test: Homogeneous groups

richness

Tukey HSD[a,b]

treatment	N	Subset 1	Subset 2
control	7	4.71	
tilled	7	5.57	
burn	7		7.71
Sig.		.545	1.000

Means for groups in homogeneous subsets are displayed.
Based on Type III Sum of Squares
The error term is Mean Square(Error) = 2.254.

a. Uses Harmonic Mean Sample Size = 7.000.

b. Alpha = .05.

This table is produced for Tukey's post-hoc tests and is called a Homogeneous Subsets table. Based on the results from the Tukey's test, treatments are put into subsets that group non-different (based on $p > 0.05$ in Tukey's test) treatments together. In this case, control and tilled are not significantly different from each other (Subset 1), and burn (Subset 2).is significantly different from both control and tilled.

Fixed vs. Random Effects ANOVA

- **Fixed effects ANOVA** (model I ANOVA)
  - Groups (treatments, levels) of the factor are assigned or controlled by the experimenter
  - Groups are specifically chosen as the only levels of interest in the study
  - E.g. plots were specifically assigned to the burn, till, control treatments, therefore treatment is a fixed factor
- **Random effects ANOVA** (model II ANOVA)
  - Groups of the factor are not assigned or controlled by the experimenter (not specifically chosen)
  - Groups are randomly selected from several possible levels of the factor

Fixed vs. Random Effects ANOVA

Fixed effects ANOVA (model I ANOVA)

- Differences observed between groups are assumed to be due to the effect of the group/treatment
- Results can be used to make predictions about these treatments in other (similar) areas/study

Random effects ANOVA (model II ANOVA)

- Differences observed between groups are due to some random effect
- Results cannot be used to make predictions about the effect of these groups in other (similar) areas/study

Assumptions of ANOVA

Assumptions deal with residuals, not the data themselves.

Residual = deviation of an observation from the sample mean, or unexplained (error) component of an observation.

$$X_{ij} - \overline{X}_i$$

i….group

X_{ij}….replicate *j* in group *i*

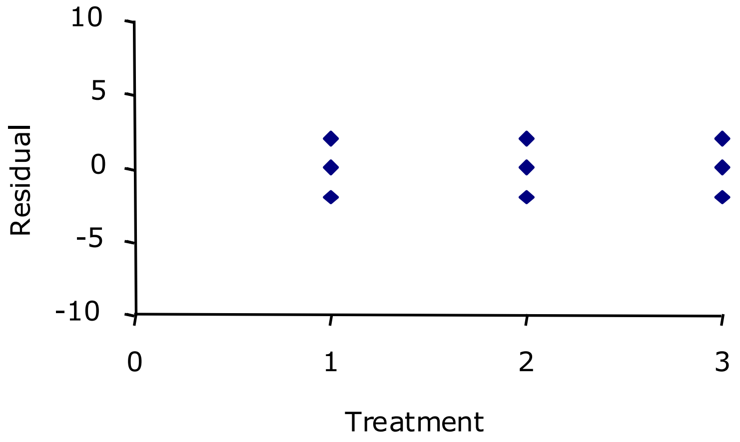
Assumptions of ANOVA

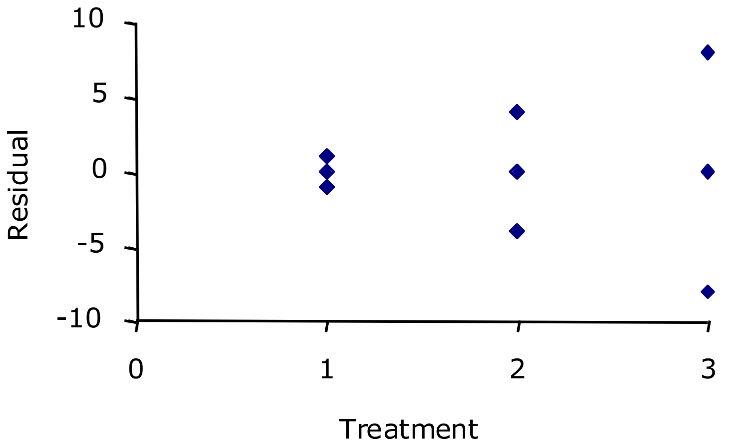
1. **Residuals come from a normal distribution**
   - Test distribution of residuals using “goodness of fit” test and probability plot/histogram
   - If not normal, first try transforming the data (e.g., log, square root), redo the ANOVA on transformed data and retest the residuals from this analysis
     - If satisfied after transformation, run the ANOVA on transformed data
     - If not satisfied after transformation, use a non-parametric test

Assumptions of ANOVA

2. Variance of residuals must be equal between groups

- Test with **Levene's test**, i.e. run ANOVA on residuals (using residuals as the dependent variable).
- Can also check graphically by looking at the spread of residuals.
- If unequal, transform **data**, redo ANOVA and recheck assumptions etc.





Assumptions of ANOVA

3. Samples from each treatment (group) must be independent from each other.

Remember:
Assumptions testing is always described in the Methods section of a scientific report!

Assumptions of ANOVA

- ANOVA is fairly robust to violations of the assumptions, especially with large sample sizes and a balanced design (equal number of replicates per group)
- Non-normality is also less problematic when the variances are equal

- If residuals are strongly non-normal and cannot be normalised and/or variances cannot be equalised through transformation

  → do a non-parametric test.

  → a non-parametric alternative for one-way ANOVA is the Kruskal-Wallis H test.

Post hoc tests and the problem of comparison-wise error

- **Type I error:** occurs if you reject the null hypothesis when it should have been not rejected.

- **Type II error:** is when a false null hypothesis is maintained/not rejected.

- Type 1 and Type 2 errors are opposites. As you reduce the likelihood of a Type 1 the chance of a Type 2 increases.

Post hoc tests tend to have a problem with Type I error

- i.e. they tend to show more significant differences than actually exist
- This occurs because every time you test the same null hypothesis (i.e. that there is no difference between groups), you have a 5% probability of rejecting the H_o for the wrong reason (i.e. H_0 is actually true)

Thus, if you are doing 10 comparisons:

- Instead of rejecting H_o when $p < 0.05$
- You only reject H_o when $p < 0.05/10$ tests
- i.e. you only reject H_o when $p < 0.005$

- Thus, this test is more conservative